

AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL

Submission: model_4

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution proves that the a_i share a common residue modulo g (the partial-progress milestone), and its overall route to the bound and construction is otherwise close to complete. The decisive issue is how it obtains coprimality: from $r \neq 0$ it infers $\gcd(r, g) = 1$, but this does not follow — a proof that no prime divides both r and g is required. The markscheme classifies exactly this step as **NEC1**, so the solution is *not* essentially complete regardless of the later work (the elimination of $k \geq 4$ is also left incomplete). It therefore scores **1** rather than a complete-with-minor-error mark.